2026-10-16 09:46:37 **You**

[] block.py
[] game_engine.py
[] main.py
[text/markdown] README.md
look through my code and fix the bug mentioned in the README where blocks stack incorrectly and the game ends even when they overlap. the blocks should end the game only when they completely miss the tower. don't change anything else, and explain what you fixed

Claude

The bug is in `drop_block()` in `game_engine.py`. The condition is inverted:

```python
is_successful_drop = overlap <= 0
```

`overlap <= 0` means the block *missed* the tower entirely, but the code treats that as a successful drop. So landing on the tower triggers game over, and missing it stacks a block. Flipping the comparison fixes it:

```python
is_successful_drop = overlap > 0
```

Here's the corrected section (this is the only line I changed):

```python
        left = max(act.x, top_block.x)
        right = min(act.x + act.width, top_block.x + top_block.width)
        overlap = right - left

        is_successful_drop = overlap > 0

        if is_successful_drop:
            trimmed_width = max(10.0, overlap)
            new_block = Block(left, act.y, trimmed_width, self.block_height, act.color,
speed=0)
            self.stack.append(new_block)
```

```
            self.score += 1
```

How it works now: `left` and `right` mark the edges of the shared horizontal region between the moving block and the top of the stack. `overlap` is the width of that region. If it's positive, the blocks share some horizontal span, so the drop is valid. The block is trimmed to the overlapping portion and added to the stack. If it's zero or negative, the blocks don't touch at all, so it's a complete miss and the game ends.

I left everything else untouched, including the `max(10.0, overlap)` minimum width, which is existing behavior and not part of this bug.

Separately, the README's folder structure section mentions `word_scramble/` and `text_box.py`, which don't match this project. That's just leftover template text and doesn't affect the game, so I didn't touch it.

2026-10-16 09:49:51 **You**

add the perfect placement feature from the readme where if a block lands almost perfectly on the tower, it should snap into place without losing any width, show a "PERFECT!" message and give bonus points, and if I get multiple perfect placements in a row, gradually restore some of the lost width. it must work with the existing scoring and restart system.

Claude

The test hit game over on the first drop because the uploaded file still contains the original inverted bug. I'll apply the earlier `overlap > 0` fix to the output file and rerun.

Perfect placement is added in `game_engine.py`, which also includes your earlier `overlap > 0` fix. Only that file changed. I tested it headlessly: perfect drops, the width restore, the streak reset, a miss, and a restart all behaved as intended.

How it works

- **Perfect detection:** a drop is perfect if the active block's `x` is within 6 pixels of the top block's `x`. The tolerance is `self.perfect_tolerance` if you want it stricter or looser.
- **Snap:** a perfect block copies the top block's `x` and full width, so nothing is trimmed.
- **Bonus points:** a perfect drop scores 1 for height plus `2 × streak` bonus, capped at 10. Your third perfect in a row is worth 1 + 6.

- **Width restore:** the first perfect in a streak only snaps. Each consecutive perfect after that restores 10 pixels, centered on the tower and capped at the original 180 base width. The block stays inside the 20px screen margins that `Block.update` enforces.
- **Streak reset:** any non-perfect landing resets the streak to 0, and so does a restart.
- **Message:** a "PERFECT!" message with the streak count (e.g. `x3`) and `+N bonus` fades out over about one second under the HUD.

Test results

Drop	Result
Imperfect (50px off)	trimmed to width 130
Perfect #1	width stays 130, bonus +2
Perfect #2	width 140, bonus +4
Perfect #3	width 150, bonus +6
Imperfect	trimmed to 110, streak reset
Complete miss	game over
Restart	score 0, streak 0, base width 180

The bonus goes into the same `self.score`, so the HUD's "Height: N" and the "Final Height" on the game-over screen now include bonus points. If you want true height and bonus points tracked separately, I can add a second counter and label.

2026-10-16 09:53:15 **You**

add the falling debris animation: whenever a block gets trimmed, the cut-off part should fall down and rotate instead of just disappearing. it must look natural and work with the camera scrolling, the other game mechanics must not break after this feature is added

Claude

The trims and cleanup work. My test stack collapsed before the camera scrolled, so I'll test that path separately.

The falling debris is added in `game_engine.py`, which still includes the overlap fix and perfect placement. Nothing else changed, and `block.py` and `main.py` are untouched. I tested it headlessly (trims on both sides, perfect drops, camera scrolling, game over, restart).

What it does

- **Piece creation:** when a block is trimmed, the overhanging piece is cut off and becomes a new `Debris` object with the same color and outline as the block. It starts exactly where the overhang was, so it looks like the block was sliced in place. If both sides are cut, you get two pieces.
- **Motion:** the piece drifts away from the tower with a small random push and accelerates downward under gravity, capped at a terminal speed.
- **Rotation:** it spins outward, so a right-side piece tips clockwise and a left-side piece counter-clockwise, at a random speed each time. It rotates around its own center.
- **Cleanup:** once a piece has fallen fully below the screen, it is removed, so the list never grows.

Interaction with the rest of the game

- **Camera scrolling:** when the tower scrolls down, the debris is shifted by the same amount, so it stays in the right place relative to the tower. In my test the debris moved by exactly one scroll step (32px) when the camera scrolled.
- **Perfect placements:** these don't trim anything, so no debris is created.
- **Game over:** pieces keep falling under the "TOWER COLLAPSED!" overlay instead of freezing.
- **Restart:** the debris list is cleared in `reset()`.
- **Scoring, speed and collision:** the debris is purely visual. It doesn't touch the score, the stack, or the drop logic.

The spin is around each piece's center, which looks natural for a quick tumble but isn't a physics simulation. If you want it to pivot around the tower's edge, I can change that.

2026-10-16 14:09:38 **You**

make the background change colours gradually as the tower gets taller, like the readme describes. after that, check that all four tasks are implemented properly and that nothing else is broken. fix any additional issues you find if there are any

Claude

Review done. I found one real bug (Space doesn't restart after game over, even though the screen and README say it does), which I'll fix with the background feature.

The background now shifts gradually as the tower grows, and all four tasks are implemented and working in the updated `game_engine.py`. I found and fixed one additional bug during the review.

Background (Task 4)

- **Colors:** the sky blends through pre-dawn blue, daytime blue, a pink-orange sunset, purple dusk, night, and finally near-black space.
- **Gradient:** it is vertical, and the top of the screen is always a few blocks "higher" in the atmosphere than the bottom, so you can see the next color coming.
- **Smoothing:** the sky eases toward the real tower height rather than jumping. In my test, no color channel changed by more than 2 units per frame.
- **Stars:** they fade in around height 34 to 48. I added this beyond the request, since a flat dark sky looked plain at the top.
- **Height used:** the background follows the actual number of blocks in the tower, not the score, so perfect-placement bonuses don't speed it up.
- **Tuning:** the colors and height thresholds are in `self.sky_keyframes` if you want to adjust them. I checked screenshots at heights 0, 10, 22, 32 and 46, and the title and HUD text stay readable.

Review of all four tasks

Task	Status
1. Overlap bug	Fixed: overlap stacks the block, a complete miss ends the game
2. Perfect placem ent	Works: snap, bonus, "PERFECT!" with streak, gradual width restore, streak re sets on a trim
3. Debris	Works: falls, spins, moves with camera scrolling, cleans up, clears on restart
4. Background	Works as described above

I also ran 300 random playthroughs checking that block widths never go below 10 or above the base width and that the score never drops below the tower height. No crashes or violations.

Bug fixed

Pressing Space on the game-over screen didn't restart. Only R or a mouse click worked, even though the screen says "Press [Space] or [R] to Play Again" and the README says the same. Space now restarts, along with R and left-click, and a restart also resets the sky.

Things I noticed but left alone

- **Score label:** because perfect bonuses go into the score, "Height: N" can be much larger than the real number of blocks. In the screenshots, a 46-block tower reads "Height: 486". Renaming the

label to "Score", or showing blocks and points separately, would fix it. Tell me which you prefer.

- **Camera scroll:** the camera moves in an instant 32px step, while the README says it "smoothly scrolls". This was in the original code and works correctly, so I didn't change it. I can add an eased scroll if you want it.
- **Minimum block width:** the original `max(10.0, overlap)` can make a block slightly wider than the overlap, so the debris piece can overlap the new block by a few pixels in rare cases. It's barely noticeable and was already in the original.
- **README:** the folder structure section still lists `word_scramble/` and `text_box.py`.